

Σ Project and Project-aware Tools

Operation	Keystroke	Function	Note
Project Dev Tools ◦ Help & Customization • Identify project • Buffer commands • Visit files ◦ Search • Dired, VC-suport, Shell • Shell, Compile, • Project commands ◦ find-file-in-project Last updated on:	Emacs provides project-aware operations natively via the project.el library. Emacs 25.1 introduced the first version of project.el. project.el was enhanced in several Emacs versions and continues to be available as a separate packing in GNU Elpa. It is possible to install the latest version of project.el in older Emacs. PEL installs and activate the following external packages when the corresponding pel-use- user-option is turned on:  find-file-in-project  pel-use-find-file-in-project Find files in project. This project is a little flaky. Prefer Σℱ Projectile  project  pel-use-project-from-elpa When set, PEL installs the latest version of project.el it finds in Gnu Elpa. Useful to get a version of project.el that is more recent than the version built in Emacs. This project still evolves and it's useful to get the latest. 2026-09-28 See also: Σℱ Projectile and Projectile vs Emacs project.el which compares Projectile to Emacs project.el built-in.		
Open this PDF file. See also: Σ Help/Info	<f11> p <f1> C-x p <f1>	(pel-help-pdf &optional OPEN-WEB-PAGE)	Open the Σ Project Tools local PDF. • With prefix argument (like C-u or M--) is used, then it opens the remote GitHub hosted raw PDF instead. • If the pel-flip-help-pdf-arg user-option is set it's the other way around.
Σ Customize PEL project support	<f11> p <f2> C-x p <f2>	(pel-customize-pel &optional OTHER-WINDOW)	Customize PEL project tools support: pel-pkg-for-project, including Σℱ Projectile and Σℱ Treemacs . • If OTHER-WINDOW is non-nil (use C-u), display in another window.
Σ Customize Emacs project support	<f11> p <f3> C-x p <f3>	(pel-customize-library &optional OTHER-WINDOW)	Customize Emacs project tools support: project and find-file-in-project • If OTHER-WINDOW is non-nil (use C-u), display in another window.
Emacs Project Support	Emacs provides the built-in project.el library. That library was first released with Emacs 25.1 and keeps evolving. The latest released version of project.el is available on Gnu Elpa as  project and may be more recent than the version you have on Emacs. This will provides extra commands on Emacs < 31. • To do that set  pel-use-project-from-elpa to t and restart Emacs. The project.el uses the following variables to control its behaviour:		
• To find a file	• find-program (␣) • project-list-files		On Emacs ≥31.1 it uses the VC-backend project-list-files function which uses the VC backend to find a file (eg. git ls-files for Git). If that is not available it uses the program identified by the find-program variable, normally set to find.
• Customize project	It's possible to store project-specific customization into an Emacs directory-local file. See Σ File/Dir Variables for more information.		
Σ Customize project's root dir-locals (Emacs≥31.1 project≥0.12)	C-x p <f4> D	(project-customize-dirlocals)	Run customize-dirlocals in current project's root. • Note that customize-dirlocal is bound to <f11> <f2> D and is available in Emacs ≥30.1 Also see Σ File/Dir Variables
• Identify project files	Emacs project maintains a list of projects inside a file identified by project-list-files user-option.		
Switch to another project (Emacs ≥ 28.1)	C-x p p	(project-switch-project DIR)	"Switch" to another project by running an Emacs command.
Search and Indetify all projects under a directory (Emacs ≥ 28.1)	C-x p <f4> A	(project-remember-projects-under DIR &optional RECURSIVE)	Index all projects below a directory DIR. • If RECURSIVE is non-nil, recurse into all subdirectories to find more projects. • After finishing, a message is printed summarizing the progress. Returns the number of detected projects.
Add a project to the project list	C-x p <f4> a	(project-remember-project PR &optional NO-WRITE STABLE)	Add project PR to the front of the project list. • On Emacs < 31: Save the result in 'project-list-file' if the list of projects has changed, and NO-WRITE is nil. • On Emacs ≥31.1: If project PR satisfies 'project-list-exclude', then nothing is done.
(Emacs≥31.1 project≥0.12)	Save the result in 'project-list-file' if the list of projects has changed. When called from Lisp, optional argument NO-WRITE non-nil means to suppress saving 'project-list-file'. Optional argument STABLE means don't move PR to the front of the project list if it's already present further down the project list.		
Remove a project from the project list (Emacs ≥ 28.1)	C-x p <f4> d	(project-forget-project PROJECT-ROOT)	Remove directory PROJECT-ROOT from the project list. • PROJECT-ROOT is the root directory of a known project listed in the project list.
Forget all known projects that do not exist anymore (Emacs ≥ 28.1)	C-x p <f4> M-d	(project-forget-zombie-projects &optional INTERACTIVE)	Forget all known projects that don't exist any more.
• Buffer Commands	See also: Σ Buffers		
Open current project's buffer	C-x p b	(project-switch-to-buffer BUFFER-OR-NAME)	Display buffer BUFFER-OR-NAME in the selected window. • When called interactively, prompts for a buffer belonging to the current project. Two buffers belong to the same project if their project instances, as reported by 'project-current' in each buffer, are identical.
Open matching buffer in another project (Emacs≥31.1 project≥0.12)	C-x p M-d	(project-find-matching-buffer)	Switch to a matching buffer in another project. • See also pel-open-file-in-other-dir bound to <f11> f M-d in Σ File-mngt
	• For most file-visiting buffers, the matching buffer is one visiting a file in the other project which has the same file name relative to the project root. • See 'project-find-matching-file' for details. • Non-file-visiting major modes may configure a different notion of matching buffer; see 'project-find-matching-buffer-function'. • When called during switching to another project, this command will detect that, and use the override. Otherwise, it prompts for the project to use from the list of known projects. • When calling from Lisp, bind 'project-current-directory-override' to a directory under the target project to preempt this prompting.		
List all buffer used by current project (Emacs ≥ 29.1)	C-x p C-b	(project-list-buffers &optional ARG)	Display a list of project buffers in a buffer named "*Buffer List". • By default, all project buffers are listed except those whose names start with a space (for internal use). • With prefix argument ARG, show only buffers that are visiting files.
Display specified buffer in some window, without selecting it	C-x 4 p C-o	(project-display-buffer BUFFER-OR-NAME)	Display BUFFER-OR-NAME in some window, without selecting it. • When called interactively, prompts for a buffer belonging to the current project. Two buffers belong to the same project if their project instances, as reported by 'project-current' in each buffer, are identical. • Uses 'display-buffer' as a subroutine, which see for how it is determined where the buffer will be displayed.
Display specified buffer in another frame.	C-x 5 p C-o	(project-display-buffer-other-frame BUFFER-OR-NAME)	Display BUFFER-OR-NAME preferably in another frame. • When called interactively, prompts for a buffer belonging to the current project. Two buffers belong to the same project if their project instances, as reported by 'project-current' in each buffer, are identical. • Uses 'display-buffer-other-frame' as a subroutine, which see for how it is determined where the buffer will be displayed.
Save some project buffers (Emacs≥31.1 project≥0.12)	C-x p C-s	(project-save-some-buffers ARG)	Like 'save-some-buffers' (bound to C-x s, see Σ File-mngt) , but only for this project's buffers.
Kill all current project buffers (Emacs ≥ 29.1)	C-x p k	(project-kill-buffers &optional NO-CONFIRM)	Kill the buffers belonging to the current project. Two buffers belong to the same project if their project instances, as reported by 'project-current' in each buffer, are identical. Only the buffers that match a condition in 'project-kill-buffer-conditions' will be killed. • If NO-CONFIRM is non-nil, the command will not ask the user for confirmation. • NO-CONFIRM is always nil when the command is invoked interactively.
• Visit Files	See also: Σ File-mngt		
Edit a file in project directory tree	C-x p C-f	(project-root-find-file filename)	Edit file. Prompts for the file name, defaulting to the root directory of the current project. (Emacs ≥ 31.1 or project 0.12)
Visit a project file Like C-x C-f but for project files only.	C-x p f	(project-find-file &optional INCLUDE-ALL)	Visit a file (with completion) in the current project. • The filename at point (determined by 'thing-at-point'), if any, is available as part of "future history". If none, the current buffer's file name is used. • If INCLUDE-ALL is non-nil, or with prefix argument when called interactively, include all files under the project root, except for VCS directories listed in 'vc-directory-exclusion-list'.
Visit a file in the project or one of its external roots (Emacs ≥ 29.1)	C-x p F	(project-or-external-find-file &optional INCLUDE-ALL)	Visit a file (with completion) in the current project or external roots. • The filename at point (determined by 'thing-at-point'), if any, is available as part of "future history". If none, the current buffer's file name is used. • If INCLUDE-ALL is non-nil, or with prefix argument when called interactively, include all files under the project root, except for VCS directories listed in 'vc-directory-exclusion-list'.

Operation	Keystroke	Function	Note	
• Search in Project Files	The first 2 commands, project-find-regexp and project-or-external-find-regexp use Xref for search and front-end (display).			
Search for regexp in project files. Show found text in Xref front end. Like <f11> g r but for project files only.	C-x p g	(project-find-regexp REGEXP)	Find all matches for REGEXP in the current project's roots (using the program identified by xref-search-program and xref-search-program-alist) and show the result with the currently selected Xref font-end (use <f11> X ? to see which one is used and <f11> X F to modify it. See Xref	
	- With C-u prefix, you can specify the directory to search in, and the file name pattern to search for. - The pattern may use abbreviations defined in 'grep-files-aliases', e.g. entering 'ch' is equivalent to '*.ch']'. - As whitespace triggers completion when entering a pattern, including it requires quoting, e.g. 'C-q<space>'.			
Search for regexp in project files and external roots (eg. libraries)	C-x p G	(project-or-external-find-regexp REGEXP)	Find all matches for REGEXP in the project roots or external roots (using the program identified by xref-search-program and xref-search-program-alist) and show the result with the currently selected Xref font-end (use <f11> X ? to see which one is used and <f11> X F to modify it. See Xref Define external roots via project-vc-external-roots-function	
Search REGEXP in project files (<i>Emacs ≥ 27.1</i>) Visit file at finding.	C-x p S	(project-search REGEXP)	Search for REGEXP in all the files of the project. Stops when a match is found, visiting matching file/line. To continue searching for the next match, type M-x fileloop-continue, or <f11> <f5> f.	
Query replace regxp in project files (<i>Emacs ≥ 27.1</i>)	C-x p r	(project-query-replace-regexp FROM TO)	Query-replace REGEXP in all the files of the project. Stops when a match is found and prompts for whether to replace it.	
	At that prompt, type a character saying what to do with the match. Type SPC or ' y ' to replace the match, DEL or ' n ' to skip and go to the next match. - For more directions, type M-x help-command at that time. If you exit the ' query-replace ', you can later continue the ' query-replace ' loop using the command M-x fileloop-continue . - PEL binds it to <f11> <f5> f (see Undo/Redo/Repeat).			
• Xref Dired				
Start Direct in project directory (<i>Emacs ≥ 28.1</i>)	C-x p d	(project-find-dir)	Start Dired in a directory inside the current project. Prompt for sub-directory of current project root. The current buffer's 'default-directory' is available as part of "future history".	
Start Direct in project root directory (<i>Emacs ≥ 28.1</i>)	C-x p D	(project-dired)	Start Dired in the current project's root.	
• VC Support	See the VC support: Xref VCS-Git xMagit , Xref VCS-Mercurial and Xref VCS-Subversion			
Open a VC Dir buffer for project. (<i>Emacs ≥ 28.1</i>)	C-x p v	(project-vc-dir)	Run VC-Dir in the current project's root. Useful to quickly open the VC Dir buffer as it does not prompt for the VC root as C-x v d does.	
• Shell Support	See also Xref start Shells/REPLs			
Start a shell in current project's root. (<i>Emacs ≥ 28.1</i>)	C-x p s	(project-shell)	Start an inferior shell in the current project's root directory. If a buffer already exists for running a shell in the project's root, switch to it. Otherwise, create a new shell buffer. With C-u prefix arg, create a new inferior shell buffer even if one already exists.	
Run sync shell command in project root directory (<i>Emacs ≥ 28.1</i>)	C-x p !	(project-shell-command)	Run ' shell-command ' in the current project's root directory. Prompts for the command.	
Run async shell command in project root directory (<i>Emacs ≥ 28.1</i>)	C-x p &	(project-async-shell-command)	Run ' async-shell-command ' in the current project's root directory.	
• EShell Support	See also Xref start Shells/REPLs			
Start an eshell in current project's root directory (<i>Emacs ≥ 28.1</i>)	C-x p e	(project-eshell)	Start Eshell in the current project's root directory. If a buffer already exists for running Eshell in the project's root, switch to it. Otherwise, create a new Eshell buffer. With C-u prefix arg, create a new Eshell buffer even if one already exists.	
• Compilation Support				
Run a compilation command in the current project root directory (<i>Emacs ≥ 28.1</i>)	C-x p c	(project-compile)	Run ' compile ' in the project root. Prompts for the command.	
Run recompile command in the current project root directory (<i>Emacs ≥ 28.1</i>)	C-x p C	(project-recompile &optional EDIT-COMMAND)	Run ' recompile ' in the project root with an appropriate buffer. With C-u prefix arg, edit the compilation command.	
• Execute Project-aware commands	The following commands allows you to run any generic command and guarantee it targets the project, even if you are currently editing a file outside of it or looking at an unrelated buffer (like *scratch*)			
Run the next typed command in the current project. (<i>Emacs ≥ 30.1</i>)	C-x p o	(project-any-command &optional OVERRIDING-MAP PROMPT-FORMAT)	Prepare execution of the next command to run inside the project. - If the command name starts with 'project-', or its symbol has property 'project-aware', it gets passed the project to use with the variable 'project-current-directory-override'. Otherwise, 'default-directory' is temporarily set to the current project's root. - If OVERRIDING-MAP is non-nil, it will be used as 'overriding-terminal-local-map' to provide shorter bindings from that map which will take priority over the global ones.	
Run the next typed command in the current project. (<i>Emacs ≥ 30.1</i>)	C-x p O	(project-prefix-or-any-command)	Run the next command in the current project. Works like 'project-any-command', but also mixes in the shorter bindings from 'project-prefix-map'.	
Project-aware command execution	C-x p x	(project-execute-extended-command)	Execute an extended command in project root. This is a project-aware M-x	
• Project command in another tab	C-x t p	(project-other-tab-command)	Run project command, displaying resultant buffer in a new tab.	
	C-b ! & D F G b c d e f g k o p r s v x	project-list-buffers project-shell-command project-async-shell-command project-dired project-or-external-find-file project-or-external-find-regexp project-switch-to-buffer project-compile project-find-dir project-eshell project-find-file project-find-regexp project-kill-buffers project-any-command project-switch-project project-query-replace-regexp project-shell project-vc-dir project-execute-extended-command		

Operation	Keystroke	Function	Note	
• Project command in another window	C-x 4 p	(project-other-window-command)	Run project command, displaying resultant buffer in another window.	
	C-b ! & D F G b c d e f g k o p r s v x C-o	project-list-buffers project-shell-command project-async-shell-command project-dired project-or-external-find-file project-or-external-find-regexp project-switch-to-buffer project-compile project-find-dir project-eshell project-find-file project-find-regexp project-kill-buffers project-any-command project-switch-project project-query-replace-regexp project-shell project-vc-dir project-execute-extended-command project-display-buffer		
• Project command in another frame	C-x 5 p	(project-other-frame-command)	Run project command, displaying resultant buffer in another frame.	
	C-b ! & D F G b c d e f g k o p r s v x C-o	project-list-buffers project-shell-command project-async-shell-command project-dired project-or-external-find-file project-or-external-find-regexp project-switch-to-buffer project-compile project-find-dir project-eshell project-find-file project-find-regexp project-kill-buffers project-any-command project-switch-project project-query-replace-regexp project-shell project-vc-dir project-execute-extended-command project-display-buffer-other-frame		
Other Tools	 find-file-in-project  pel-use-find-file-in-project			
Find file in project	<f11> p f f	(find-file-in-project-by-selected &optional OPEN-ANOTHER-WINDOW)	Prompt for file glob, search in project for the file.  This code is weak and breaks often. Prefer the project commands above and § 3 Projectile as well as other PEL specific file management commands that provide lots of functionality. See § File-mngt and the language specific pages.	